

Line	Code	Explanation
1	.MODEL SMALL	Declares memory model = SMALL → program has one data segment and one code segment , each max 64 KB.
2	.STACK 64	Reserves 64 bytes of stack space for temporary storage (used during function calls, interrupts, etc.).
3	.DATA	Begins the data segment , where variables are defined.
4	DATA1 DB 52H	Defines a byte variable DATA1 = 52H (82 in decimal) .
5	DATA2 DB 25H	Defines a byte variable DATA2 = 25H (37 in decimal) .
6	SUM DB ?	Defines a byte variable SUM , uninitialized (? means its value is not set yet). It will store the result.
7	.CODE	Begins the code segment , where instructions (program logic) start.
8	MAIN PROC FAR	Declares the main procedure , entry point of the program. FAR means it can be called from another segment (uses both CS:IP).
9	MOV AX, @DATA	Loads the starting address of the data segment into register AX. (@DATA gives segment address of .DATA).
10	MOV DS, AX	Initializes the Data Segment Register (DS) so that the program can access data variables correctly.
11	MOV AL, DATA1	Moves the value of DATA1 (52H) into AL register (the lower byte of AX).
12	MOV BL, DATA2	Moves the value of DATA2 (25H) into BL register.
13	ADD AL, BL	Adds BL to AL → $AL = AL + BL$ → result: $52H + 25H = 77H$.
14	MOV SUM, AL	Stores the sum (77H) back into memory variable SUM .
15	MOV AH, 4CH	Prepares for program termination by setting AH = 4Ch , the DOS interrupt function code for "Exit to DOS" .
16	INT 21H	Executes DOS interrupt 21H → function 4Ch → returns control to the operating system.
17	MAIN ENDP	Marks the end of the MAIN procedure .
18	END MAIN	Tells the assembler that program execution starts at MAIN .